heads h_i and h_j respectively:

$$s^{(m,n)}_{(h_i,h_j)}(p) = CosineSim\left(\mathrm{vec}\big(A^m_{h_i}(p)\big), \mathrm{vec}\big(A^n_{h_j}(p)\big)\right) \tag{1}$$

3.2.2. Step 2: Average similarity score across prompts

We aggregate over prompts to obtain an average similarity scores between the two heads (h_i and h_j):

$$\bar{s}^{(m,n)}_{(h_i,h_j)} = \frac{1}{|\mathcal{P}|}\sum_{p\in\mathcal{P}} s^{(m,n)}_{(h_i,h_j)}(p) \tag{2}$$

3.2.3. Step 3: Best-match head in the pair refit

By repeating Step 2 (Eq. 2), We compare head h_i against all possible candidate heads h_j in the pair refit ($\{1,\ldots,H^n\}$). Then, we choose the best matching head based on their *average similarity scores*:

$$s^{(m,n)}_{h_i} = \max_{h_j\in\{1,\ldots,H^n\}} \bar{s}^{(m,n)}_{(h_i,hj)} \tag{3}$$

We interpret $s^{(m,n)}_{h_i}$ as the *stability* of head h_i with respect to the single refit n. For instance, a representative plot for an 8-layer architecture Fig. A.5 visualizes the stability of all heads from an anchor refit with with respect to other individual pair refits.

3.2.4. Step 4: Stability over all refits

Finally, we average the single-refit stability (Eq. 3) across all pair refits to obtain the overall stability of head h_i across all N refits:

$$S^{(m)}_{h_i} = \frac{1}{N-1}\sum_{n\neq m} s^{(m,n)}_{h_i} \tag{4}$$

Interpretation: $S^{(m)}_{h_i}$ indicates the propensity of anchor head h_i to be matchable to a head in the *same layer* with a *similar attention-head representation* across independently initialized refits. In this way, we obtain a principled, permutation-invariant notion of attention head stability that obviates the need for Hungarian matching (Ainsworth et al., 2023).

Next, we can repeat this procedure for every head in the anchor refit m.

3.3. Cross-layer best-match variant

We also consider a relaxation of the § 3.2 procedure that drops the layer constraint when identifying a best match. For an anchor head (l, h_i) in refit m, the candidate set includes all heads from all layers (rather than only layer l) in the pair refit n. That is, h_j can be any head belonging to any layer in *pair* refit n. The rest of the method remains identical.

4. Results

In this section we present the experimental results that underpin our main contributions. We discuss ten key results that probe cross-seed stability from complementary perspectives. For brevity, we report results only for representative architectures in the main body; the complete set of similar assessments for other model architectures is provided in App. B.

4.1. Head-wise and layer-wise stability

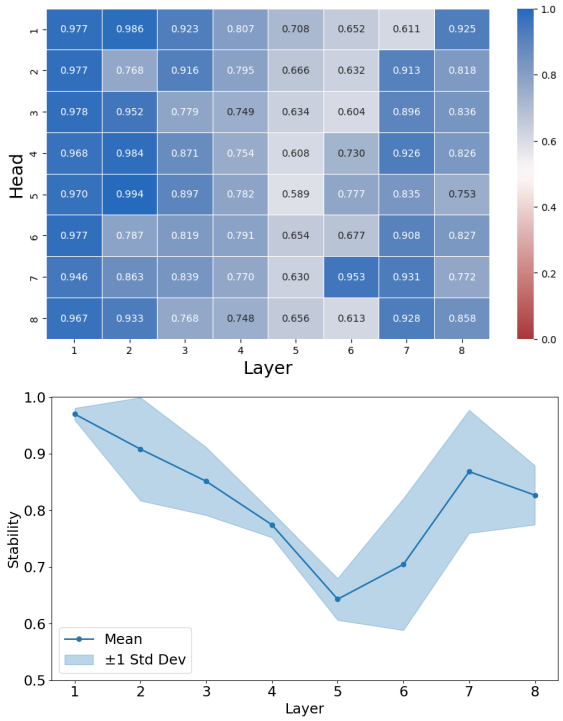

Figure 2. **Only some mid-layer attention heads persist across refits.** For each head h_i from layer l in an anchor model, We find its best-matching counterparts in the same layer l across other refits and use $S^{(m)}_{h_i}$ from Eq. 4 as the head's stability (top). This plot thus summarizes inter-seed stability for all the heads of the anchor. Consistently, initial and final layers are more stable than middle layers (bottom).

Head-wise stability: Using the metric defined in § 3.2 and a 100-prompts set, we compute the stability of every head in a single **anchor** instance drawn from different refits of a given architecture. The results allow for a bird's eye view of cross-seed stability: many refits are compared against one anchor within the same architecture. For an 8-layer, 8-head MLP architecture (Fig. 2), we consistently observe a higher cross-seed stability in the earlier and later layers, with a pronounced dip in stability measures corresponding to attention heads in the middle layers of the architecture. At the head level, profiles vary: some layers are uniformly stable (or uniformly unstable) across seeds, whereas others

exhibit patterns with variance.

Layer-wise stability: From the ensuing head-wise estimates, we compute a layer-wise mean by averaging the stability of all heads within a layer. Here, $h_i \in H$, where H denotes the set of all heads in layer l, and m denotes the anchor refit.

$$S_l^{(m)} = \frac{1}{|H|} \sum_{h_i \in H} S_{h_i}^{(m)} \tag{5}$$

Fig. 2 highlights the mid-depth instability. For example, $S_5^{(m)}$ (layer 5) drops to ≈ 0.70, down from 1.0 in layer 1. This finding highlights the inherent challenge in identifying truly universal circuits in the intermediate layers of transformer models. Comparable figures for other architectures are provided in the § B.1.

4.2. Cross-layer best-match stability

Relaxing the layer constraint: As noted in § 3.3, we also compute a cross-layer best-match stability by allowing each anchor head to match with any head of any layer in the pair refit. As expected, the resulting stability is slightly higher that the stability calculated with layer constraints (§ 4.1), simply due to the larger candidate pool. However, overall trend was similar. Representative plots for a few of the architectures are provided in the App. B.2.

Alignment Map & Middle-layer dispersion: To visualize *"Which layer do the best matches come from?"*, we construct a row-normalized alignment matrix (H) (shown in Fig. 3) with entries ($H_{i,j}$). The y-axis denotes the anchor head's layer index (i); the x-axis denotes the layer index (j) of the best-matched heads. Rows sum to 1. The heatmap exhibits strong diagonal dominance, that is, most heads match to a similar layer in paired refits, confirming a strong layer wise correspondence across seeds.

Compared to early and late layers, middle layers show a broader, flatter row distribution (*lower kurtosis*), with substantial off-diagonal mass into neighboring layers. This cross-layer spillover is also consistent with mid-depth instability and indicates a less sharply defined layer-to-layer correspondence for those heads.

4.3. Within-layer uniqueness of attention heads

Method: For anchor refit m, head h_i in layer l is compared to all other heads in the same layer using attention score matrices over the prompt set $\mathcal{P}$ (as in § 3.2). Additional details are provided in App. A.6. The mean similarity of an attention head to its peers quantifies the commonness (non-uniqueness) of the head.

Result & Interpretation: As shown in Fig. 4 and § B.3, mid-layer heads are most unique; early/late layers are much more prototypical—mirroring the head stability pattern described in § 4.1.

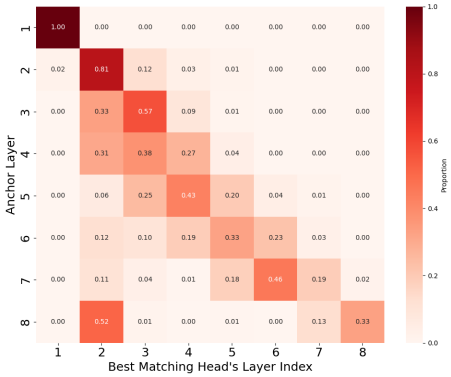

Figure 3. **Middle layer attention heads exhibit off-layer matching across re-fits.** Cross-layer best match alignment heatmap shows strong diagonal dominance with heads typically matching to same-depth layers but broader off-diagonal spread in mid-layers.

4.4. Factors promoting attention head instability

Building on the layer-wise stability analysis in § 4.1, we sought simple, model-internal correlates of head stability. We focused on parameters that directly shape the attention score matrices, namely query, key, value, and output projection matrices, as well as the pre-attention LayerNorm. Thus, we conducted an exploratory analyses (CKA-based summaries, diagnostic visualizations, and layer-wise statistics) on these objects. Two of the consistent signals that emerged were related to (i) ℓ_2 norm of the query weights and (ii) prompt length.

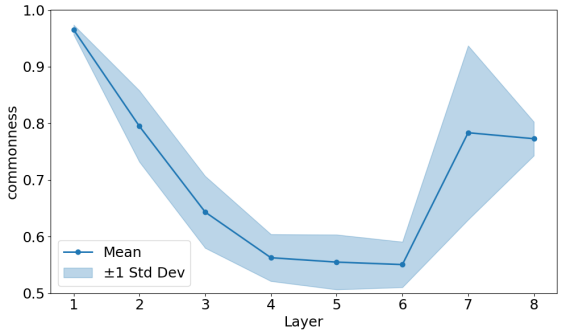

Figure 4. **Middle layer attention heads are most unique.** Attention heads are most redundant in early and late layers of GPT-like models, while being quite distinct in the intermediate layers.

4.4.1. Correlation between Query-weight norm and layer-wise stability

We compared layer-wise stability from § 4.1 to the mean (ℓ_2) Frobenius norm of the attention heads' query-weight matrices of the same layer. We observe a clear **negative correlation**—layers with larger mean query-weight norm tend

to exhibit lower stability. As shown in fig. B.5, this trend holds across architectures. This observation is especially intriguing in light of prior work linking parameter (ℓ_2) norm growth to training instabilities and related failure modes (Lyle et al., 2024; Wortsman et al., 2024). We reproduce these findings in the context of circuit identification and interpretability.

4.4.2. Effect of prompt length on stability

We next recomputed the layer-wise stability using multiple prompt sets whose length spanned 5-50 tokens per prompt. We find that stability decreases sharply with the prompt length, indicating that longer prompts induce greater token-token interaction variability and reduce cross-seed alignment (Fig. 5).

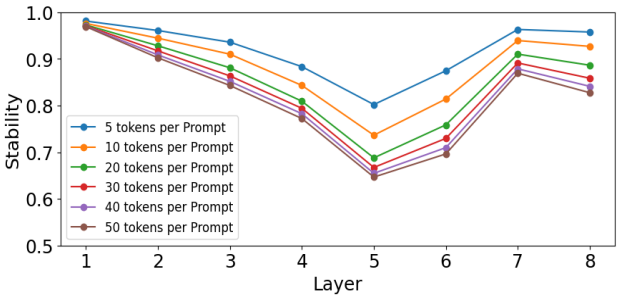

Figure 5. Instability of attention heads from middle transformer layers increases with increasing prompt length.

4.5. Stability comparison: Adam vs AdamW

Motivation: If cross-seed instability partly arises from uncontrolled norm growth, as explored in § 4.4.1, then explicitly regularizing norms during pretraining may provide a means to improve head stability. Prior work shows that AdamW (decoupled weight decay) aids convergence, generalization, and norm control (Loshchilov & Hutter, 2019; Zhang et al., 2018; D'Angelo et al., 2024). Given these findings, it is possible that AdamW also improves the universality of circuit identification across model refits. We therefore trained mirror sets of refits that are architecturally identical to the Adam runs but use AdamW as the optimizer.

Results: Studying Figures 6 and App. B.6, we see that AdamW yields a clear increase in head seed-stability for deeper MLP models (8- and 12-layer) and for all attention-only architectures. In contrast, head seed-stability gains for shallow MLP models (2- and 4-layer) are small, possibly due to limited norm growth at low depth.

Performance Parity: Despite the stability gains, replacing Adam with AdamW typically leaves validation perplexity nearly unchanged, indicating that there are no performance tradeoffs when using AdamW. See § B.7 and (Ginsburg et al., 2019).

Mechanistic check: To verify that AdamW controls norm growth (Loshchilov & Hutter, 2019), we compared the mean (ℓ_2) norm of attention-head output activations between matched Adam and AdamW refits. Using AdamW consistently suppresses activation norms, supporting the hypothesis that controlling norm growth improves the cross-seed stability. (See Fig. B.6.1).

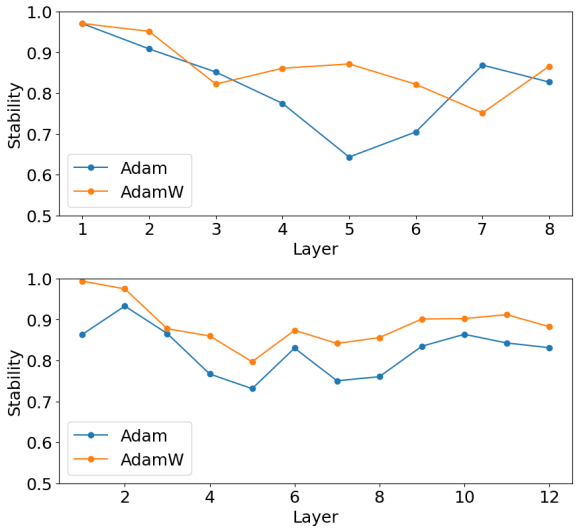

Figure 6. AdamW substantially improves attention head seed stability with no significant difference in performance. **Top**: 8-layer, 8-head, MLP architecture. **Bottom**: 12-layer, 12-head, MLP (GPT2-small) architecture.

4.6. Most- and least-stable layers

4.6.1. Stability gap between the most- and least-stable layers

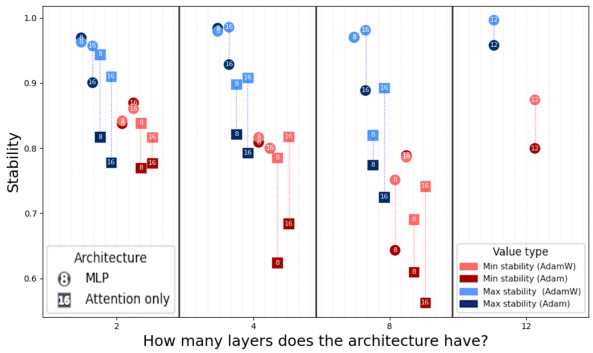

Figure 7. $S_{l_{\max}}$ (blue) and $S_{l_{\min}}$ (red) stability vs. depth. Circles: MLP; squares: attention-only. Dark shades: Adam; light: AdamW. Numbers indicate heads per layer. **Architectures trained using AdamW exhibit increased attention head stability.**

Definition: For each of the 26 architectures (§ 3.1), we compare layer-wise stability S_l (§ 4.1) and identify the most-stable layer $l_{\max}$ and least-stable layer $l_{\min}$. We visualize